Experiment 13: Parse Tree Generation using CFG in Python

1. Aim

To write a Python program to generate a parse tree for a given sentence using a context-free grammar.

2. Procedure

1. Import `CFG` and `ChartParser` from the `nltk` module.
2. Define the production rules of the context-free grammar using `CFG.fromstring()`.
3. Initialize `ChartParser` with the defined grammar.
4. Parse the tokenized input sentence to generate its syntactic tree.
5. Display the generated parse tree structure.

3. Program

```
import nltk
from nltk import CFG

grammar = CFG.fromstring("S -> NP VP\nNP -> Det N\nVP -> V NP\nDet -> 'the'\nN -> 'boy' | 'book'\nV -> 'reads'")

parser = nltk.ChartParser(grammar)
sentence = "the boy reads the book".split()

for tree in parser.parse(sentence):
    print("Generated Parse Tree:", tree)
```

4. Output

```
Generated Parse Tree: (S (NP (Det the) (N boy)) (VP (V reads) (NP (Det the) (N book))))
```

5. Result

Thus, the Python program was successfully executed to generate a parse tree for a given sentence using a context-free grammar.